

Dimensionality Reduction Balancing Local and Global Order Preservation

Davi Pereira-Santos¹ , Tácio T. A. T. Neves² , Andre C.P.L.F. de Carvalho¹ , and F. V. Paulovich³ 

¹Instituto de Ciências Matemáticas e de Computação, Universidade de São Paulo, Brazil

²Departamento de Ciências Exatas, Universidade Federal da Paraíba (UFPB), Rio Tinto - PB, Brazil

³Eindhoven University of Technology: Eindhoven, Noord-Brabant, NL

Abstract

Dimensionality Reduction (DR) is one of the most popular information visualization techniques for mapping multidimensional data into 2D or 3D layouts, allowing the creation of visual representations that depict the underlying structure of the original data. The literature reports several methods that differ in the type of structure they intend to preserve. The two most common are pairwise distances between data instances, used by globally focused techniques, and the neighborhood of each element, used by locally focused techniques. Although local techniques have become prevalent (e.g., t-SNE), interpreting the resulting layouts requires caution, as some apparent macrostructures may be artifacts produced by an inherent deficiency of the technique or an unfortunate choice of parameters (e.g., perplexity). On the other hand, global methods excel at preserving overall structure (e.g., PCA, MDS), but they often struggle to reveal clusters of similar instances, map non-linear data manifolds, or preserve local distance rankings. Although some techniques claim to preserve both local and global structures, prior work has shown that they are limited in achieving a good balance between the two. This paper presents tau-map, a novel DR technique that addresses these issues by directly optimizing the order of elements via XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX weighted and unweighted components of a standard measure, the Kendall τ correlation index. The proposed measure provides precise values about layout quality while balancing local and global structure preservation and inhibiting visual artifacts. We formally introduce its differentiable version and the corresponding parametric DR algorithm, with out-of-sample capabilities and time complexity suitable for high-dimensional data. Empirical results against state-of-the-art techniques show that tau-map attains the best trade-off among existing techniques while also allowing users to control, possibly in real time, whether to emphasize local or global structure preservation.

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CCS Concepts

• **Computing methodologies** → Collision detection; • **Hardware** → Sensors and actuators; PCB design and layout;

1. Introduction

Dimensionality Reduction (DR) techniques are among the most popular visualization strategies for the analysis of multidimensional data, being ubiquitous to different domains, from explainable AI applications [?] to the study of genes [?] and proteins [?]. In general lines, DR techniques aim to reduce the multidimensional data to 2 or 3 dimensions, allowing for the creation of a visual layout that represents some underlying structure present in the original data. In the current literature, there are different taxonomies to classify the existing DR methods. Probably the most accepted one splits the techniques into global and local strategies [?]. While global techniques, like MDS [Tor65], PCA [Jol02], or PLMP [PSN10] focus on retaining in the visual layout the pairwise distances between the data instances, the local ones, like t-SNE [VdMH08], ISOMAP [TdSL00], or LSP [PNML08] focus on preserving local neighborhoods.

Although local techniques have become prevalent nowadays, given their capacity to untangle high non-linear manifolds into (usually) well-defined clusters in the visual representation, interpreting the resulting layouts requires caution since some of the presented visual structures may be visual artifacts. For instance, for some of the most popular techniques, the distances between clusters are meaningless [?]. Even clusters' density cannot be inferred from the visual representation. On the other hand, global methods are good at preserving such "overall" structures, but they usually suffer in revealing the existing groups of similar instances, which is generally a relevant trait for the general users of DR techniques [?]. Although some techniques, like UMAP [MHM18] or LSP [PNML08] claim to be able to preserve local and global structures, in practice, this showed not to be true [?], and none of these techniques allows for the users to control the trade-off between global and local preservation. Another problem is the lack of some

way to determine the reliability level of each layout element or even if the nearest neighbor is really the most similar to the center element amongst the immediate neighborhood.

These (XXXX) hindrances/deficiencies/flaws are addressed in the present paper by a novel mapping technique, called *SORTBmap*, that optimizes a Soft Order-weighted Ranking measure based on the Kendall Tau-b index for *Balanced mapping* with respect to the local-global trade-off. It is an application of the sort-ness measure, an order-aware quality index proposed in a previous work [?]. The measure is further developed here as a generalization able to cover the local-global trade-off through an explicit control mechanism that avoids creating undesirable artifacts. *SORTBmap* optimizes a new cost function which balances local and global components, enabling the map to preserve both neighborhood and overall dataset structure. This cost function outweighs the most usual function found in the literature to measure neighborhood preservation, the trustworthiness [?], amplifying it to consider not the ranking of intrusive points only, but instead calculate the agreement level between original and layout order, with the advantage of replacing the hard neighborhood boundary by a soft unrestricted one [?]. To the best of our knowledge, this is the first mapping technique to directly optimize the neighborhood order, and the first to optimize the larger structures order while keeping local features.

We compare our method against the state-of-the-art [?] DR techniques through qualitative and quantitative experiments to show that *SORTBmap* presents the best trade-off between local and global preservation if compared to the most popular techniques while also allowing for user intervention to define the focus of the preservation. In summary, our main contributions are:

- A differentiable replacement for the Kendall τ -b rank correlation index, which is widely used as a standard index of ranking agreement; and, despite having superior features, it is overlooked in favor of Spearman ρ due to the existence of a soft replacement for the latter;
- A novel DR cost function that addresses the limitations in the concept of trustworthiness, namely its lack of a smooth derivative and its loose matching between the original data and layout orderings;
- A technique which can be used as a robust parameterless one at first, with the option to allow users to explicitly control the balance between local and global trustworthiness in a more strict sense, up to the ordering, for confident interpretation;
- The balancing mechanism avoids the need to choose the number of neighbors, although we provide a parameter to set a smooth neighborhood when the user wants to ensure most of the weight is given to them for a detailed case by case visual analysis of pivot elements that depends on relative proximity;
- A more reliable approach that avoids some of the artifacts generated by the most popular local DR techniques;
- XXXXXXXXXXXXXXXXXX our reductor is able to perform out-of-sample projections

FERNANDO, o que vc acha de falar que o kappa proposto aqui é muito mais proximo do numero de vizinhos, pois é livre do problema da perplexity (nao eh vinculado a um raio de valor absoluto ou densidade)? eh independente do dataset e

realmente significa um numero soft de vizinhos voltado apenas para casos excepcionais do usuario, em vez de ser um parametro que afeta drasticamente o resultado como na tsne

The remainder of this paper is organized as follows: Sec 2....

2. Related Work

As previously discussed, the usual taxonomy divides the existing Dimensionality Reduction (DR) techniques into global and local strategies. There is also the idea of topological preservation, such as the Neighbor-Joining technique [?] or the TopoMap [BFD*18], but we refrain from discussing them since they are more related to graph theory than distance preserving DR field.

Global techniques are the ones where the focus is on preserving the pairwise distance between points in the reduced space, given their counterparts' distances in the original space. Multidimensional Scaling (MDS) [Tor65] is one of the most popular techniques among the global ones. MDS analytical solution based on the spectral decomposition of the pairwise distance matrix, and if the original space is a Euclidean space, it is guaranteed that MDS produces the best possible preservation of the original distance in a Frobenious norm sense [?]. Principal Component Analysis (PCA) [Jol02] similarly uses the spectral decomposition but on top of the covariance matrix instead of the distance matrix. PCA produces the same results as MDS if the multidimensional space is the Euclidean space [?].

PCA and, especially, MDS can be slow depending on the size of the dataset and its dimensionality since the distance and the covariance matrixes grow quadratically based on the size number of dimensions of the data, respectively. So, some approximations have been suggested, to speed up global techniques. Examples are the Landmarks MDS [DST04], Pekalska [PdRDK99], and PLMP [PSN10]. In this process, these techniques use the out-of-sample concept where a sample of the data is projected, and the remaining points are interpolated using different strategies, using linear system solutions [PSN10, PdRDK99] to neural networks [PdRDK99]. Another interesting example of global techniques is the Random Projection (RP) [Vem05]. RP uses a fairly simple but effective idea of computing a quasi-orthogonal random matrix which, when multiplied by the original data, projects it to the visual space. RP is definitely very fast, but the results are not the best.

Compared to the local techniques, the layouts created by global techniques are more reliable in representing the relationships between groups and outliers since the overall distance is also preserved in the layouts produced by such techniques. Inside the group of global techniques, the linear techniques compose an interesting sub-group of techniques. Since such techniques derived the 2 or 3 visual dimensional linearly combining the original features in the data, such sub-groups of techniques are easier to understand than the non-linear techniques. PCA and MDS are good examples of such techniques and their interpretation can be easily attained through biplot-like representations [GH95] where the importance of the original features to the definition of the visual layout can be, up to an extent, understood [CMN*16].

We could mention other examples of linear techniques, such as

XXXX [?], the literature is quite vast (see [?, ?] for good surveys). Despite the existing differences between such techniques, the underlying concept is always the same: try to approximate the pairwise-distance of the original space into the visual space.

Global techniques are easier (or possible) to interpret, and the global structures are quite reliable; however, they often fail to represent the manifolds presented in the original space, and they are usually mixed on the final layout. The result is the separation of existing clusters of similar instances is usually impaired [?]. Local techniques are usually much better at capturing or uncovering such clusters.

The general idea of local techniques is the preservation of local neighborhoods, ignoring the large distance that usually distorts the DR layouts [?]. The most prominent example is the t-SNE [VdMH08] technique. t-SNE is an adaptation of SNE [HR02], and instead of preserving distance between instances, it aims at preserving the probabilities of an instance being among the nearest neighbors of the other instances.

3. Method

XXXXXX dá pra passar parte disso para conclusão se faltar assunto lá ou em outras seções; e aqui entrar direto nas definições

We propose a method based on the simple and essencial principle that order matters when comparing objects, be them single elements or large clusters. A comparison may seek to answer which of two objects are the most similar to a central object of interest taking into account all the others that may exist in between or around them. In visualization, the simple presence of objects not directly related to the investigation in course can be misleading due their effect on the human intuition. Therefore, assessing the order of neighbors, implicitly provides a measure of how well preserved the layout is when compared to the original data. In a previous work we called this concept *sortedness*. It is based on the order of the unrestricted neighborhood of each element and is intended to reflect how well the data is organized, e.g., how clean or correct it is regarding precision/recall, intrusions/extrusions, trustworthiness/continuity. Such properties are dependent on (and less constrained than) ordering and, therefore, are included by transitivity when order is considered. Additionally, we do not consider the local-global trade-off a zero-sum game. As will be shown, e.g., in the tricoil experiment (??) when a technique does well on the local scale, it is not at the expense of the global scale. For this reason, our balancing parameter is intended to stay in the middle of both extremes as trying to weight more to one side or the other risks neglecting one aspect without benefits to the other.

3.1. Measure

The core of the technique is to compare the agreement level between the order of neighbors in the original data and their respective order in the evolving layout. This is done in three components whose generalized mean we call *balanced sortedness*. Each of them is calculated by differently-focused applications of the kendall τ variant b correlation coefficient: τ weighted by distance ranks on the original data; τ weighted by distance ranks on the map; and traditional (unweighted) τ . Kendall τ - b correlation coefficient, or rank

correlation index, was our choice against Spearman ρ for some important reasons. It can be computed directly from observations without first converting them to ranks, it has a clear intuitive and operational interpretation [Noe81], better statistical properties, and is more robust than Spearman ρ [Noe81] the most common alternative.

We separate parameters that are part of the target (hard) measure from the optimizer hyperparameters as the choice of an adequate optimizer is still an open question and not relevant for the final result, provided it is able to maximize the measure. We opted to introduce some parameters in the measure formulation for the sake of completeness and to cover unforeseen settings. However, none of them is intended to be necessarily changed by the user or tuned to improve results. They are available for specific cases when the user needs to do a detailed investigation that depends on defining how many neighbors are expected to be reliable or if the focus should be on precision or recall as follows:

- $0 \leq \alpha \leq 1$ - Balance between original ($\alpha = 0$) and map ($\alpha = 1$) rankings when defining weights; this focuses more on avoiding, respectively, extrusions and intrusions; the default value is $\alpha = 0.5$ as this is the more conservative setting, e.g., one usually wants the map to equally keep correct neighbors and avoid false neighbors; this is analogous to have a balance continuity-trustworthiness, although with the added feature of enforcing order; changing α has low impact on the layout as optimizing in one direction mostly improves the other as a side-effect.
- $0 \leq \beta \leq 1$ - Balance between local (weighted) and global (unweighted) sortedness; the default value is $\beta = 0.5$ as this is the most conservative choice, i.e., local and global structures are considered equally relevant; departing from this value is not expected to bring much benefit, risking to harm the other side of the balance.
- $1 \leq \kappa < |X|$ - Proxy to the original parameter σ of the Half Gaussian distribution to enable the choice of how many neighbors should account for 90% of the weight; $\kappa = 5$ is suitable for most applications, provided that $0 < \beta < 1$ (i.e., both local and global components are active), preferably $\beta \approx 0.5$; after κ neighbors the weights approach zero rapidly - the default value implies a total of only 13 neighbors with significant weights ($v(r) < 10^{-5} \forall r > 13$) and $v(r_5)/v(r_1) \approx 0.34$;
- $-\infty < \gamma < \infty$ - Level of tolerance for the spread between different component values; negative γ values increasingly penalize the measure when any of the component values approaches -1 ; this is intended to attenuate the effect of a very good component compensating a very bad one; the default is $\gamma = -1$ which is equivalent to use the geometric mean (moderate penalty for spreads); $\gamma = 0$ and $\gamma = 1$ are equivalent to, respectively, the arithmetic mean (no penalty) and quadratic mean (moderate reward); while the default is probably suitable for all foreseeable applications, we opted to introduce the parameter for completeness and generality of our formulation.

A reasonable order of priority for them, sorted from more relevant to less relevant to the user is $\kappa \succ \alpha \succ \gamma \succ \beta$.

The balanced sortedness of a pair dataset/layout is defined in

Equation (1).

$$\Lambda_\lambda(\mathbf{r}, \hat{\mathbf{r}}, w) = 2s - 1 \quad (1)$$

$$s = \begin{cases} l_{\mathbf{r}, \hat{\mathbf{r}}}^{z-\beta+1} l_{\hat{\mathbf{r}}, \mathbf{r}}^{z-\beta}, & \theta = -1 \\ \left[(z - \beta + 1) l_{\mathbf{r}, \hat{\mathbf{r}}}^{\theta+1} - z l_{\hat{\mathbf{r}}, \mathbf{r}}^{\theta+1} + \beta g^{\theta+1} \right]^{\frac{1}{\theta+1}}, & \text{otherwise.} \end{cases}$$

$$z = \alpha\beta - \alpha$$

$$l_{\mathbf{r}, \hat{\mathbf{r}}} = \frac{\tau_\lambda(\mathbf{r}, \hat{\mathbf{r}}, w) + 1}{2} \quad (2)$$

$$g = \frac{\tau_\lambda(\mathbf{r}, \hat{\mathbf{r}}, \mathbb{Z}_+^2 \mapsto 1) + 1}{2} \quad (3)$$

β is a parameter to balance between the global component g defined in ??, and the local components $l_{\mathbf{r}, \hat{\mathbf{r}}}$ and $l_{\hat{\mathbf{r}}, \mathbf{r}}$ defined in ?. Two local components exist due to the existence of two weighting possibilities which could dependent on the ranking of distances according to the original data or according to the map.

Let $w(r_i, r_j) = v(r_i) + v(r_j)$ be a weighting function representing the importance of a given pair of ranking positions according to v in Equation (4) based on the Cauchy distribution; given two rankings $\mathbf{r}, \hat{\mathbf{r}}$, the differentiable surrogate τ_λ of a weighted adaptation of the Kendall τ correlation index is shown in ?.

$$v(r) = \frac{\kappa}{3\pi[(r-1)^2 + (k/3)^2]} \quad (4)$$

In the ?? we explain how to make τ differentiable so that an optimization can be done in tractable time through gradient descent.

3.2. Surrogate Function

Kendall τ - b correlation coefficient, or rank correlation index, was our choice against Spearman ρ for some important reasons. It can be computed directly from observations without first converting them to ranks, avoiding dependence on sophisticated solutions like the soft rank proposed by [BTBD20], and, therefore, allowing easy customization and integration in specific algorithms. Furthermore, it has a clear intuitive and operational interpretation [Noe81], better statistical properties, and is more robust than Spearman ρ [Noe81]. Additionally, the differentiability of the measure was identified as a gap in the literature where we could contribute with a soft replacement for τ . We could not find any such an index in the literature, and even a very recent article explicitly states that the Kendall τ correlation ranking index is non differentiable and should be replaced by alternatives such as l_p -distance regularizers [WK22].

The only parameter of the surrogate function is the regularizer λ . A tie occurs when two positions have the same value in any of the rankings.

the condition of agreement/disagreement is quantified by the difference between the distance deltas approximately/smoothly binarized through the application of the hyperbolic tangent. Let $\lambda \in \mathbb{R}^+$ be a regularization term, the differentiable step function $s: \mathbb{R} \mapsto [-1; 1]$ shown in Equation (5) is intended to squash a signed value without losing differentiability.

$$s_\lambda(v) = \frac{2}{(1 + e^{-2v/\lambda})} - 1 \quad (5)$$

Therefore, beyond the need to have the parameter λ as a regularizer, it also enables the user of the measure to make it more stress-oriented or more order-oriented.

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ainda falta derivar

3.3. Computational Cost

The measure is calculated for each point in the dataset. Since it depends on Kendall τ which has $\mathcal{O}(n \log n)$ time complexity, one would expect $\mathcal{O}(n^2 \log n)$ for the global component. However, the differentiable soft step cannot take advantage of the Knight's algorithm ?? which is based on mergesort to count agreements and disagreements by transitivity. An approximate variant of the surrogate function would be needed to enable the reductor to take advantage the of algorithm lower complexity. $\leftarrow \leftarrow \leftarrow \leftarrow$ **future work XXXXXXXX** In this study, we resort to another kind of approximation which is to sample a constant number K of neighbors independently of the dataset size. This brings the complexity down to $\mathcal{O}(n^2)$ as $K \log K$ amounts to a constant factor when compared to n^2 . The same is valid for the local components, as the nearest neighbors are ordered by the *selection algorithm* [?] which has time complexity $\mathcal{O}(k \log k)$. In the latter case, there is no impact on the measure accuracy as the weights are already null after r_k .

Next we summarize the implementation-oriented parameters. They are not related to the nature of what is being measured, nor to the optimization algorithm. They provide the choice of reducing computational cost possibly at the expense of the reliability of the soft function as a surrogate of the target measure:

- $1 \leq k < |X|$ - Hard limit for the number of neighbors with an effective weight for calculations; k is usually sufficiently greater than κ to avoid disposing relevant neighbors; it is automatically set to the rank position where the weights start to have values lower than 10^{-5} ; the value of k can be set by the user in exceptional cases, if for any reason they need it; the higher the better and slower, but after some point, increasing it has no effect;
- K - Number of elements to sample for the global component calculation; the higher the better and slower;

3.4. Embedding

XXXXXX precisaria checar se embedding é a palavra correta; a cada década muda a terminologia graças ao tio hinton XXXXXXXx

We implemented an embedding neural network as a proof of concept that the differentiable balanced sortedness is optimizable for a DR effective to overcome the issues listed in the introduction. It is a traditional Multilayer Perceptron trained to predict the coordinates of the output elements. This means our reductor is able to perform out-of-sample projections. The loss function is the negation of the surrogate measure calculated on the predicted coordinates against the original data.

- L - Tuple with the number of hidden neurons in each layer; in our preliminary tests $L = \langle 50 \rangle$ showed satisfactory results for all datasets;

- σ - Activation Function - Our preliminary tests showed that a ReLU function might perform better for lower reductions in dimensionality than tanh possibly due to the more linear mapping between input and output;
- E - Epochs of optimization; in many datasets, a few epochs was enough for good results, although dozens or more epochs help in stabilizing towards a more definitive map;

3.5. Optimizer

In our preliminary tests with a few optimizers showed they perform well regardless of tuning parameters. RMSProp [?] was particularly fast to reach good solutions. RMSProp keeps a moving average of the square of gradients, dividing each gradient by the root of this average. Its parameters are presented as follows:

- minibatch size - Size of each chunk of points provided at each optimization step at once;
- learning rate - Rate at which the weights are updated in the direction of the gradient;
- momentum - Parameter to set the cumulative effect of previous updates on the future ones;
- weight decay - Regularization penalizing large weight values;
- other - Other less relevant parameters are omitted for brevity.

por ultimate opt? ver se teve melhores resultados

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XXXXXXXXXXXXXXXXXXXXXXXXXXXXCCCCC-
CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC

A definição de uma função de ponderação é importante para a componente local da medidade; vizinhos a uma curta distancia contam mais

a formula ?? é construido tendo por base a formula do kendall tau

In the limit, the proposed surrogate function tends to the b variant of Kendall τ when $\beta = 0$ as shown in Equation (6).

$$\begin{aligned}
 \lim_{\lambda \rightarrow 0} \tau_{\lambda}(\mathbf{r}, \hat{\mathbf{r}}, w_0) &= \lim_{\lambda \rightarrow 0} \frac{\sum_{i < j} \delta_{\lambda, ij} \hat{\delta}_{\lambda, ij} 2n^{-1}}{\sqrt{\sum_{i < j} |\delta_{\lambda, ij} 2n^{-1}| \cdot \sum_{i < j} |\hat{\delta}_{\lambda, ij} 2n^{-1}|}} \\
 &= \frac{\sum_{i < j} b_1 b_2 2n^{-1}}{\sqrt{\sum_{i < j} |b_3| \cdot \sum_{i < j} |b_4| \cdot 2n^{-1}}} \\
 &= \frac{\sum_{i < j} a(b_1, b_2) - \sum_{i < j} d(b_1, b_2)}{\sqrt{[\binom{n}{2} - \sum_{i < j} t(b_3)] \cdot [\binom{n}{2} - \sum_{i < j} t(b_4)]}} \\
 &= \tau(\mathbf{r}, \hat{\mathbf{r}})
 \end{aligned} \tag{6}$$

where $b_1, b_2, b_3, b_4 \in \{-1; 0; 1\}$ represent the squashed values $\delta_{\lambda, ij}, \hat{\delta}_{\lambda, ij}$ resulting from the limit demonstrated by Equation (7), in order that the number of agreements, disagreements, and ties can be counted by functions $a, d: \mathbb{R}^2 \rightarrow \{0; 1\}$ and $t: \mathbb{R} \rightarrow \{0; 1\}$

shown in equations 8 and 9.

$$\begin{aligned}
 \lim_{\lambda \rightarrow 0} s_{\lambda}(v) &= \lim_{\lambda \rightarrow 0} \frac{2}{1 + e^{-2v/\lambda}} - 1 \\
 &= \lim_{x \rightarrow \inf} \begin{cases} 2/(1 + e^x) - 1, & v < 0 \\ 2/(1 + e^0) - 1, & v = 0 \\ 2/(1 + e^{-x}) - 1, & v > 0 \end{cases} \\
 &= \begin{cases} 1, & v > 0 \\ 0, & v = 0 \\ -1, & v < 0 \end{cases}
 \end{aligned} \tag{7}$$

$$a(\delta, \hat{\delta}) = \begin{cases} 1, & \delta = \hat{\delta} \\ 0, & \text{otherwise} \end{cases} \quad d(\delta, \hat{\delta}) = \begin{cases} 1, & \delta \neq \hat{\delta} \\ 0, & \text{otherwise} \end{cases} \tag{8}$$

$$t(\delta) = \begin{cases} 1, & \delta = 0 \\ 0, & \text{otherwise} \end{cases} \tag{9}$$

A tie occurs when two positions have the same value in any of the rankings.

aqui começa uma tentativa de mostrar que lambda grande faz a tangente hiperbolica se aproximar da funcao linear, que é a ausencia de ranking e influencia total da distancia

Conversely, higher λ values provide a less smooth s_{λ} . Increasing λ makes s_{λ} approximate a linear function $f(v) = cv, c \in \mathbb{R}$ which makes ?? to be equivalent to the stress function if, instead of ranks, distances to neighbors were adopted.

prova mais formal disso:

Despite the similarity with Kruskal stress formula 1 for large λ values, our measure still differs from stress in this extreme because the difference between distances is multiplied by another difference between distances, instead of being used directly. Nonetheless, due to the exposure of the difference without softening in this case, we speculate λ to be analogous to a balance between exact ordering (as defined by Kendall index) and a MDS-like behavior.

Therefore, beyond the need to have the parameter λ as a regularizer, it also enables the user of the measure to make it more similar to stress or more order-oriented.

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XXXXXXXXXXXXXXXXXXXX

ongoing work

XXXXXXXXXXXXXXXXXXXX

Justificativas para balancear:

- medida única ponderada (i.e. sem balancear) é afetada por n
- medida única ponderada precisa de cauda mais longa do que uma distribuição de probabilidade pode fornecer, ou seja, acaba precisando usar a sequência harmônica (que não soma 1); o fato dela não ser distribuição, a torna menos rigorosa para fins de justificativa, ou e.g. amostragem de vizinhos
- quando se adota cauchy (igual t-sne) a distribuição tem cauda

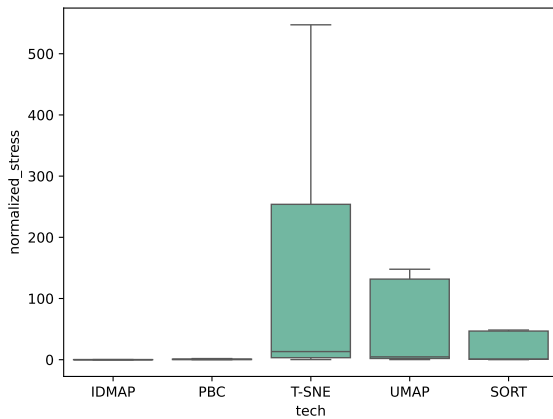


Figure 1: Normalized stress.

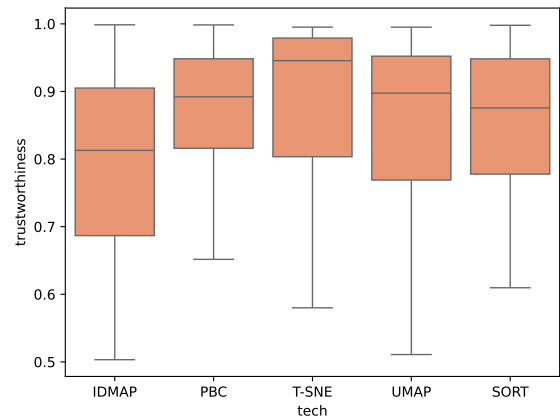


Figure 2: Trustworthiness.

longa, mas já não serve para global, pois para κ grande, os pesos ficam muito homogêneos (o que seria equivalente a não ponderar); então, cauchy é boa, mas requer balancear com uma outra medida, a global.

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4. Results and Evaluation

As Figuras ilustram os resultados das métricas: stress normalizado, trustworthiness, continuity, sortedness cau e sortedness kendalltau. Eu removi os outliers. Os valores de stress continuam estranhos. Estou chamando de SORT a projeção com a Sortedness por enquanto. Os gráficos estão considerando as últimas projeções enviadas pelo Davi, após a busca pelos parâmetros ótimos.

XXX acrescentar $\beta = 0$ $\beta = 1$ como técnicas; testar se $\alpha = 0$ inverte trust-continuity; acrescentar $\beta = 0.5$ como medida

4.1. How to use effectively

In their article, the authors present simple examples to demonstrate the capabilities and limitations of maps generated using t-SNE. Here, we will replicate these examples, comparing them with the results presented by SORT. We replicated the code provided by the authors in Python, generated datasets for each example, and created maps using t-SNE and SORT techniques. In all images, we used the parameters that created the best visually evaluated t-SNE map: perplexity = 50 and step = 5000. Except for Figure 10, which had the best visual evaluation with perplexity = 100.

Cluster sizes in a t-SNE plot mean nothing

This experiment analyzes the visual map generated when the two groups have different standard deviations, causing a different intra-group spread. Figure 6 illustrates the result.

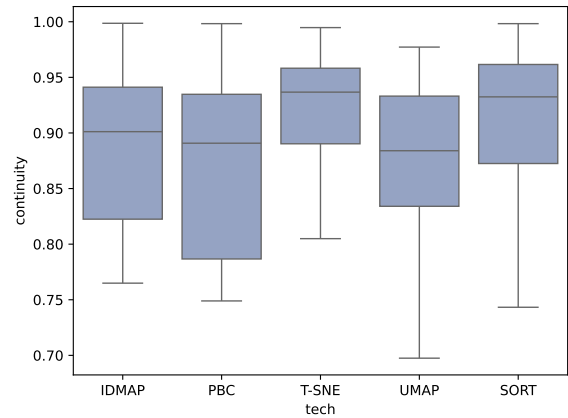


Figure 3: Continuity.

As mentioned in the original work, the groups on the map created by t-SNE appear to be the same size. On the map created by SORT, the groups reflect the difference in size.

Distances between clusters might not mean anything

In the following experiment, an analysis of the intercluster distance is made. For the first analysis, three groups were created with 50 instances each, given a Gaussian distribution. The results is shown in Figure 7

The parameters used were identical, but the t-SNE-generated visual map did not exhibit the same distance relationship between groups as observed in the original article. In contrast, SORT produced a visual map that maintained a similar distance relationship.

The same experiment was then replicated, increasing the number of instances in each group from 50 to 200, as shown by Figure 8. Although SORT did not yield a favorable outcome, the initial study upheld t-SNE's consistent behavior and equal group distribution.

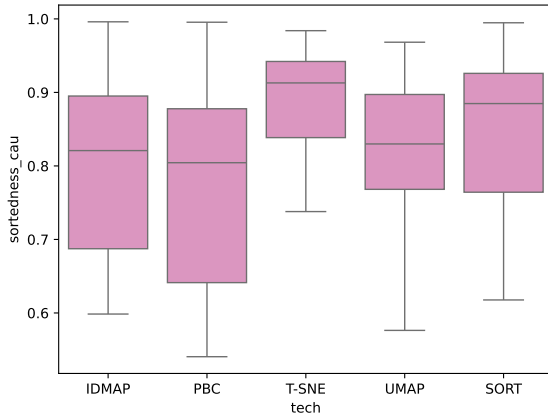


Figure 4: Sortedness Cau.

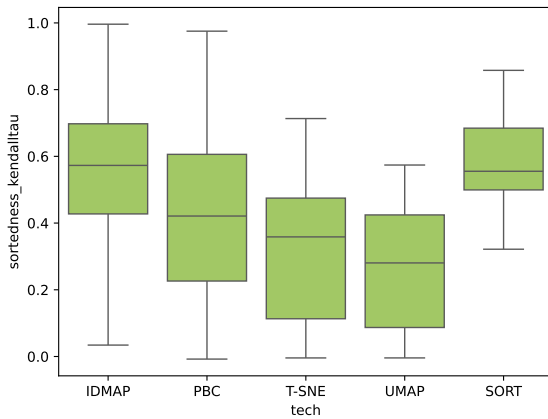


Figure 5: Sortedness Kendalltau.

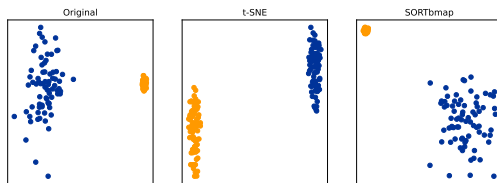


Figure 6: The second experiment examines visual map differences due to varying standard deviations, leading to distinct intragroup spreads.

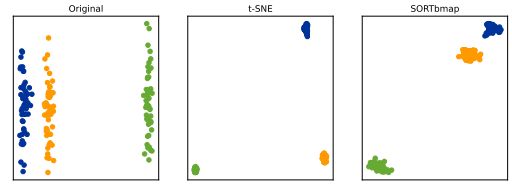


Figure 7: Experiment that analyzes intercluster distance. Three sets of Gaussian distributions, each containing 50 instances.

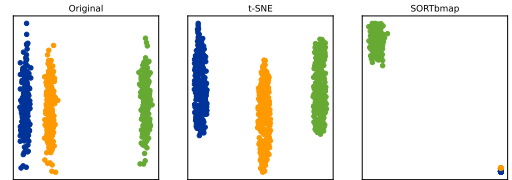


Figure 8: Experiment that analyzes intercluster distance. Three sets of Gaussian distributions, each containing 200 instances.

The relative positions of clusters in a t-SNE plot are not always reliable indicators of their actual relationships.

Random noise doesn't always look random

You can see some shapes, sometimes

For topology, you may need more than one plot

5. Discussions and Limitations

6. Conclusions and Future Work

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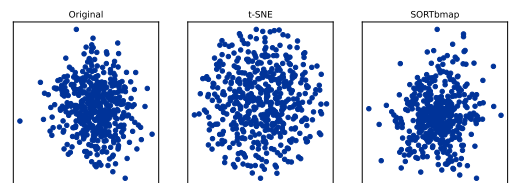


Figure 9: ...

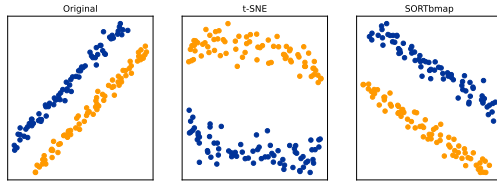


Figure 10: ...

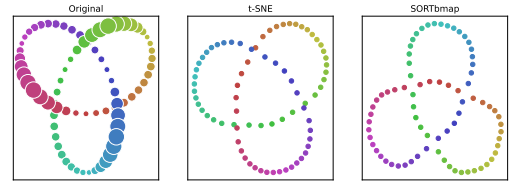


Figure 13: ...

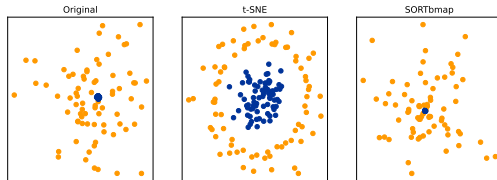


Figure 11: ...

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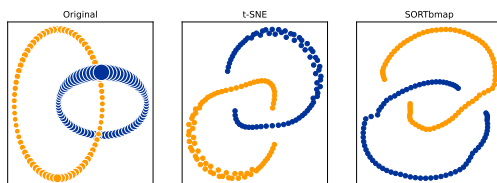


Figure 12: ...

$$w_i^{(t)} = w_i^{(t-1)} + \Delta w_i$$

$$\Delta w_i = \eta(y - \hat{y})x_i$$